

CERTIFIED CIGAR SOMMELIER (CCS®)

MODULE 9 TEACHING MANUAL

SETTLING TIME SCIENCE, THERMODYNAMICS & PEAK-FLAVOUR METHODOLOGY

Status: Advanced Core Module – Mandatory

Applies to: CCS® Level II & ACS® Level III

Difficulty Level: High

Estimated Study Load: 3 Hours

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15. MODULE SUMMARY

MODULE PURPOSE

This module establishes the scientific framework behind:

- cigar settling behaviour,
- peak-flavour development,
- stabilization science,
- and the thermodynamic mechanisms governing cigar recovery after:
 - production,
 - shipping,
 - storage,
 - and environmental transition.

The module formalizes the:

Peak-Flavour Methodology

developed within the:

Cigar Peak-Flavour System®.

Candidates will learn:

- why cigars continue evolving after production,
- why moisture equilibrium alone does not guarantee peak flavour,
- how tobacco architecture influences stabilization,
- how thermal shock affects aromatic coherence,
- and how multiple independent settling systems interact simultaneously.

A candidate who does not master this module cannot competently diagnose:

- cigar maturity,

- smoking readiness,
- storage behaviour,
- or stabilization quality.

LEARNING OUTCOMES

By the end of this module, the candidate shall be able to:

1. Explain cigar settling using thermodynamic terminology
2. Distinguish axial equilibrium from radial equilibrium
3. Explain why moisture coherence does not equal flavour coherence
4. Describe hygroscopic stabilization in long-filler cigars
5. Explain how temperature alters moisture migration kinetics
6. Distinguish factory maturation from humidor reconditioning
7. Explain ammonia dissipation scientifically
8. Diagnose thermal shock destabilization
9. Explain why certain tobacco types require lower EMC targets
10. Apply the V23 Peak-Flavour readiness methodology professionally

SECTION 1

THE CIGAR AS A THERMODYNAMIC SYSTEM

1.1 The Cigar as a Moisture-Reactive Biological Structure

A premium cigar is:

- hygroscopic,
- anisotropic,
- thermally reactive,
- oxygen-diffusion limited,
- and chemically unstable over time.

A cigar continuously exchanges:

- moisture,
- heat,
- volatile compounds,
- and gases

with its surrounding environment.

Unlike static consumer products:

cigars remain chemically active long after production.

1.2 Why Cigars Continue Evolving After Production

Even after rolling and boxing:

- moisture gradients remain unstable,
- volatile compounds continue redistributing,
- residual ammonia continues dissipating,
- oils continue migrating,
- and aromatic integration continues developing.

This means:

a cigar can be physically finished but chemically unfinished.

1.3 The Cigar as a Directional Structure

Premium long-filler cigars are not homogeneous cylinders.

They are:

- layered,
- directional,
- and structurally anisotropic systems.

They are composed of:

- multiple filler leaves,
- aligned axially,
- wrapped concentrically,

- and compressed under variable density.

This creates:

- directional airflow,
- directional moisture migration,
- and directional thermal propagation.

The cigar therefore stabilizes differently:

- longitudinally (axially),
and
- radially (core-to-wrapper).

SECTION 2

AXIAL MOISTURE COHERENCE

2.1 Definition

Axial moisture coherence refers to:

the equalization of moisture distribution between the cap and foot along the cigar's longitudinal structure.

This process governs:

- combustion stability,
- ignition continuity,
- thermal propagation,
- and flavour progression.

2.2 Why Axial Coherence Matters

When axial moisture is unstable:

- combustion zones behave inconsistently,
- smoke density fluctuates,
- pyrolysis becomes unstable,
- and flavour architecture fragments.

Typical symptoms include:

- tunneling,
- uneven flavour transitions,
- intermittent bitterness,
- inconsistent draw resistance,
- and muted aromatic development.

2.3 Why Long-Filler Cigars Settle Slowly

Long-filler cigars stabilize slowly because:

- tobacco is hygroscopic,
- oxygen diffusion is restricted,
- leaf packing is dense,
- and moisture migration is diffusion-limited.

Additional slowing factors include:

- thick ligero content,
- broadleaf wrappers,
- oily tobaccos,
- large ring gauges,
- and dense bunching.

Axial stabilization is therefore:

not a simple humidification event.

It is:

a slow diffusion process occurring through layered biological material.

2.4 Axial Equilibrium Does Not Equal Peak Flavour

A cigar may achieve axial moisture coherence while still lacking:

- aromatic integration,
- smoke softness,

- flavour layering,
- and combustion harmony.

Therefore:

axial coherence is necessary but insufficient for peak flavour.

SECTION 3

RADIAL HUMIDOR EQUILIBRIUM

3.1 Definition

Radial equilibrium refers to:

the moisture equalization between the cigar core and the surrounding humidor environment.

This process governs:

- moisture adaptation,
 - burn speed,
 - smoke density,
 - and combustion temperature regulation.
-

3.2 Core Moisture vs Ambient Moisture

A cigar entering a humidor may contain:

- more moisture than ambient,
- less moisture than ambient,
- or unstable internal gradients.

The cigar therefore begins exchanging moisture with the environment.

This exchange is governed by:

- vapor pressure gradients,
- absolute moisture content,
- diffusion resistance,
- temperature,

- and leaf density.
-

3.3 Drying vs Humidifying Behaviour

Radial settling behaves differently depending on:

the direction of moisture movement.

Drying-Down Behaviour

When the cigar core contains more moisture than ambient:

- lower-temperature air may still extract moisture effectively, because:
- colder air contains lower absolute water content at the same RH.

Therefore:

lower temperatures do not always slow drying.

Humidifying-Upward Behaviour

When the cigar core is drier than ambient:

- colder temperatures slow moisture uptake, because:
- diffusion kinetics decline,
- and available water energy decreases.

This is why:

upward humidification slows dramatically below approximately 19°C.

3.4 Practical Implications

Radial equilibrium strongly affects:

- combustion speed,
- smoke texture,
- thermal stability,
- and harshness perception.

However:

radial equilibrium alone does not guarantee mature flavour.

SECTION 4

AMMONIA DISSIPATION

4.1 Origin of Ammonia

Ammonia is a natural byproduct of tobacco fermentation.

It emerges through:

- nitrogen decomposition,
- protein breakdown,
- and microbial fermentation activity.

Improperly stabilized cigars may continue releasing ammonia during smoking.

4.2 Sensory Effects of Ammonia

Excess ammonia commonly presents as:

- throat irritation,
- nasal sharpness,
- chemical harshness,
- bitterness,
- and unstable retrohale behaviour.

These effects are frequently misdiagnosed as:

- blend strength,
 - nicotine intensity,
 - or pepper complexity.
-

4.3 Why Cuban Cigars Require Longer Ammonia Settling

Many Cuban cigars:

- receive limited post-roll aging,

- are shipped relatively fresh,
- and continue stabilizing after boxing.

As a result:

Cuban cigars often require extended ammonia dissipation periods.

Typical full stabilization may require:

- approximately 52–58 weeks.

4.4 Dominican and Nicaraguan Behaviour

Dominican cigars frequently undergo:

- longer post-roll aging periods,
- reducing ammonia burden before shipment.

Nicaraguan cigars:

- often contain denser leaves,
- are frequently rolled wetter,
- and may still require additional settling after procurement.

SECTION 5

FACTORY MATURATION VS HUMIDOR RECONDITIONING

5.1 A Critical Distinction

The Peak-Flavour Methodology distinguishes:

factory maturation

from:

humidor reconditioning.

These are not the same process.

5.2 Factory Maturation

Factory maturation occurs:

- before shipment,
- during post-roll aging,
- during warehousing,
- and during controlled storage.

This phase stabilizes:

- ammonia,
- volatile compounds,
- oils,
- and initial blend integration.

5.3 Humidor Reconditioning

Humidor reconditioning occurs:

- after procurement,
- within the smoker's environment,
- under new humidity and temperature conditions.

This process restores:

- combustion coherence,
- aromatic openness,
- smoke softness,
- and flavour integration.

A mature cigar may still require:

substantial humidor reconditioning.

5.4 Why Broadleaf-Heavy Cigars Require Longer Marrying

Broadleaf-heavy cigars stabilize slowly because:

- wrappers are thicker,
- oils are denser,
- moisture buffering is stronger,

- and combustion is naturally slower.

These cigars frequently require:

- significantly longer blend marrying periods.

SECTION 6

TEMPERATURE SHOCK RECOVERY

6.1 Definition

Temperature shock refers to:

rapid environmental thermal exposure during shipping or storage.

Examples include:

- winter freezing,
- overheated cargo containers,
- cold cargo holds,
- airport storage,
- and summer transport spikes.

6.2 What Thermal Shock Destabilizes

Temperature shock affects:

- volatile aromatic distribution,
- oil mobility,
- micro-moisture gradients,
- combustion behaviour,
- and aromatic coherence.

The cigar may become:

- muted,
- fragmented,
- harsh,

- or aromatically “closed.”

6.3 Recovery Periods

Shock Severity Typical Recovery

Mild	2–4 weeks
Moderate	6–8 weeks
Severe	10–12 weeks

Broadleaf-heavy cigars may require even longer recovery periods.

SECTION 7

TOBACCO-SPECIFIC THERMODYNAMIC BEHAVIOUR

7.1 Why Different Tobaccos Stabilize Differently

Not all tobacco behaves identically.

Different leaf families possess different:

- density,
- oil content,
- hygroscopicity,
- combustion behaviour,
- and thermal buffering.

These differences materially affect:

- ideal EMC,
- settling speed,
- combustion temperature,
- and aromatic expression.

7.2 Yamasá Tobacco Behaviour

Yamasá tobacco often expresses:

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- dark mineral character,
- espresso notes,
- earth,
- cocoa,
- and black pepper.

Because of:

- denser structure,
- mineral-rich soils,
- and stronger combustion behaviour,

Yamasá frequently performs best at:

slightly lower EMC targets
than softer Dominican tobaccos.

Excess moisture commonly causes:

- muted espresso character,
- heavy smoke texture,
- combustion drag,
- and reduced mineral clarity.

7.3 Broadleaf Behaviour

Broadleaf wrappers:

- absorb and retain moisture strongly,
- slow combustion,
- increase smoke density,
- and resist rapid environmental equilibration.

They therefore:

- require longer settling,
- tolerate slightly drier targets,
- and punish over-humidification severely.

SECTION 8

THE PEAK-FLAVOUR METHODOLOGY

8.1 Why Peak Flavour Is Multidimensional

Peak flavour does not occur when:

- moisture alone stabilizes.

Instead:

peak flavour emerges only when multiple independent stabilization systems converge simultaneously.

8.2 The Five Readiness Systems

The methodology models five independent readiness systems:

1. Axial moisture coherence
 2. Radial humidor equilibrium
 3. Ammonia dissipation
 4. Blend marrying / sensory maturation
 5. Temperature shock recovery
-

8.3 Peak-Flavour Readiness Logic

Final peak readiness is defined as:

the latest remaining stabilization requirement among all five systems.

Therefore:

a cigar is not considered peak-ready until all systems converge simultaneously.

8.4 Different Systems Use Different Clocks

Axial Coherence

Primarily credited from:

- stable humidor rest.

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Ammonia Dissipation

Primarily credited from:

- production date.

Blend Marrying

Primarily credited from:

- production date,
- post-roll aging,
- and controlled maturation history.

Thermal Shock Recovery

Primarily credited from:

- stable post-shock humidior recovery.

Radial Equilibrium

Calculated from:

- current core moisture,
- ambient RH,
- temperature,
- and blend architecture.

SECTION 9

PROFESSIONAL DIAGNOSTIC APPLICATIONS

Scenario 1 - Fragmented Broadleaf Cigar

Professional Diagnosis

- axial coherence may already be achieved,
but
 - blend marrying remains incomplete.
-

Scenario 2 - Winter Shipment Muting

Professional Diagnosis

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- thermal shock destabilization,
not necessarily:
 - poor storage practice.
-

Scenario 3 - Chemically Sharp Cuban Cigar

Professional Diagnosis

- incomplete ammonia dissipation,
not necessarily:
 - excessive strength.
-

Scenario 4 - Over-Conditioned Yamasá Blend

Professional Diagnosis

- likely over-conditioned,
resulting in:
 - suppressed mineral clarity,
 - muted espresso character,
 - and excessive smoke heaviness.
-

SECTION 10

COMMON MYTHS (AND WHY THEY ARE WRONG)

Myth 1

“All cigars stabilize at the same RH.”

Why This Is Incorrect

Different tobacco architectures possess:

- different hygroscopic behaviour,
 - different combustion characteristics,
 - and different EMC requirements.
-

Myth 2

“Moisture equilibrium means the cigar is ready.”

Why This Is Incorrect

Moisture stability does not guarantee:

- flavour marrying,
- aromatic integration,
- or ammonia dissipation.

Myth 3

“A cigar that burns well is mature.”

Why This Is Incorrect

Combustion stability does not guarantee:

- flavour coherence,
- smoke softness,
- or aromatic completeness.

Myth 4

“Temperature only affects beetle activity.”

Why This Is Incorrect

Temperature fundamentally alters:

- diffusion kinetics,
- moisture movement,
- stabilization speed,
- and volatile compound behaviour.

SECTION 11

EXAMINATION ALIGNMENT

Candidates must be able to:

- distinguish axial from radial equilibrium,
- explain why cigars continue evolving after production,
- explain ammonia scientifically,
- describe thermal shock effects,
- explain why broadleaf-heavy cigars settle slowly,
- explain why peak flavour requires convergence of multiple systems,
- and defend tobacco-specific humidity targeting scientifically.

Common Examination Traps

Candidates often:

- confuse moisture balance with maturity,
- confuse harshness with strength,
- assume all tobacco behaves similarly,
- or ignore thermal history.

Professional analysis must remain:

- thermodynamic,
- evidence-based,
- and multidimensional.

SECTION 12

ORAL DEFENSE STATEMENTS

Candidates should be able to defend the following statements orally:

1. A cigar may be moisture-stable but not flavour-stable.
2. Peak flavour requires convergence of multiple independent settling systems.
3. Broadleaf-heavy cigars stabilize more slowly because of density and oil behaviour.
4. Thermal shock can destabilize an otherwise mature cigar.
5. Ammonia dissipation is not identical to flavour marrying.
6. Different tobacco types require different EMC targets.

7. Axial equilibrium and radial equilibrium are physically distinct phenomena.
 8. Temperature alters stabilization kinetics, not just humidity perception.
-

MODULE SUMMARY

This module establishes that:

- cigar settling is governed by thermodynamics,
- moisture coherence alone does not guarantee peak flavour,
- different tobacco families stabilize differently,
- thermal history materially affects aromatic performance,
- and professional cigar diagnosis requires multidimensional scientific analysis.

This methodology therefore replaces:

simplistic RH folklore

with:

structured thermodynamic interpretation.

Without mastery of this module:

advanced cigar maturation analysis becomes unreliable.